Caring Types: How levels of retention, curation and preservation drive transparent repository metadata

International Digital Curation Conference 2025

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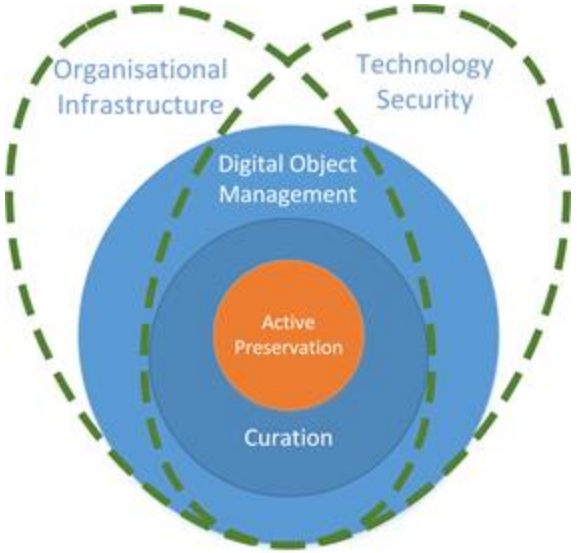
CoreTrustSeal: Context

- International, community based, non-governmental, and non-profit organization promoting sustainable and trustworthy data infrastructures
- Founded in 2017
- Born from the Data Seal of Approval and the World Data System (WDS)
- Certification based on community-agreed requirements
  - 16 requirements covering Organisational Infrastructure, Digital Object Management, and Information Technology & Security
  - Meant as a core level certification



CoreTrustSeal 2023-2025

- Organisational Infrastructure
  - Mission & Scope (R01)
  - Rights Management (R02)
  - Continuity of Service (R03)
  - Legal & Ethical (R04)
  - Governance & Resources (R05)
  - Expertise & Guidance (R06)
- Digital Object Management
  - Provenance & Authenticity (R07)
  - Deposit & Appraisal (R08)
  - Preservation Plan (R09)
  - Quality Assurance (R10)
  - Workflows (R11)
  - Discovery & Identification (R12)
  - ReUse (R13)
- Information Technology & Security
  - Storage & Integrity (R14)
  - Technical Infrastructure (R15)
  - Security (R16)

CoreTrustSeal: Levels of Curation

- Developed through open public feedback
- Arrived at four levels:

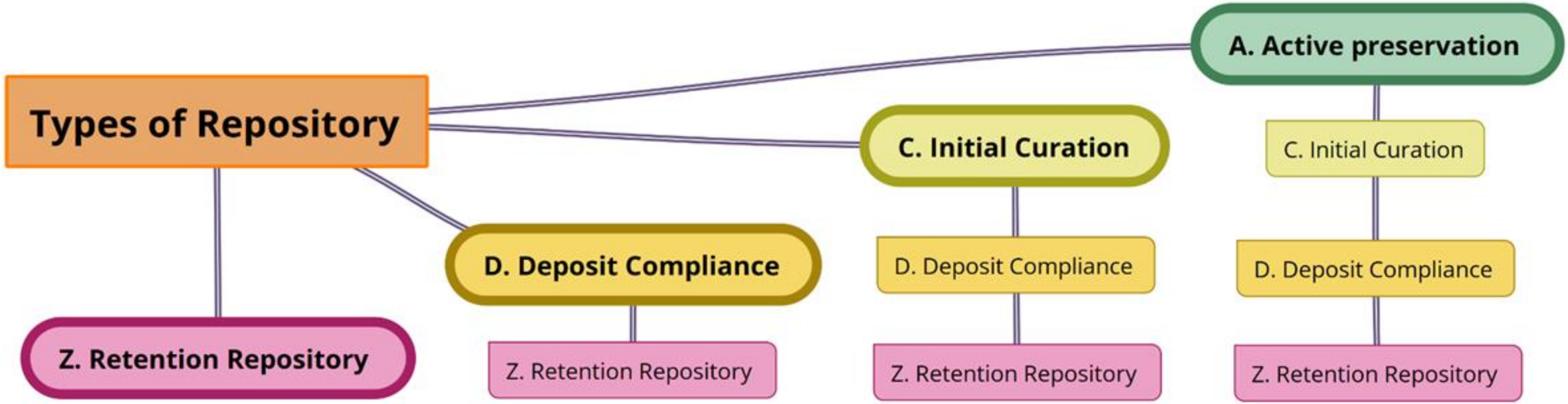
  Z. Level Zero
    - Content distributed as deposited. Unattended deposit-storage-access

  D. Deposit Compliance
    - Non-compliance triggers rejection or requires initial curation

  C. Initial Curation
    - Repository acts as required to meet defined criteria

  A. Active Preservation
    - Long-term responsibility to act as required to ensure reuse
- Repositories may offer **Z**, **D** and/or **C**, but must offer **A** to be in scope for CoreTrustSeal

Typology


Types of Repository
A. Active preservation
C. Initial Curation
C. Initial Curation
D. Deposit Compliance
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Z. Retention Repository
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Levels of Curation – Implied Metadata

- Working paper explored:
  - Metadata that could be associated with each level
  - How this metadata could support curation and preservation actions and improve transparency
  - How this could be stated at the repository level and at the object level
    - Curation and Preservation levels:
      - Repository level: All of the levels of curation and preservation provided
      - Object level: All of the levels of curation and preservation this object received

Levels of Curation – Implied metadata

	Repository Level	Object Level
Z. Retention Repository	Standard/Minimum/Maximum **Retention Periods** (Time)	Minimum Retention Period: Time Maximum Retention Period: Time Retention Start Date: YYYY-MM-DD
	Exceptions to retention periods applied by the repository.	Exception to retention periods for this specific object.
D. Deposit Compliance	Documented criteria that a digital object should meet at the point of deposit. Including Semantic (metadata, documentation, rights), technical (format), quality (formal data quality, scientific quality, ethical).	Link to repository level criteria in place when the object was deposited.
	Outcomes. All potential outcomes of the Deposit Compliance assessment.	Information about the outcome for this specific object.
	Exceptions. All possible exceptions to the deposit compliance criteria and outcomes.	Information about any exceptions applied to this specific object.
C. Initial Curation	Documented criteria that a digital object should meet at the end of the initial curation process. Including Semantic (metadata, documentation, rights), technical (format), quality (formal data quality, scientific quality, ethical).	Link to repository level criteria in place when the object was initially curated.
	Exceptions. All possible exceptions to the criteria and outcomes'	Information about any exceptions applied to this specific object.
A. Active Preservation	Standard/Minimum/Maximum Preservation Periods (Time).	Preservation Period (Time) and Start (Date) for this specific object.
	Exceptions to Preservation periods applied by the repository.	Exception to preservation periods for this specific object.
	Preservation Action triggers applied by the repository	History of preservation actions applied to the specific digital object as part of provenance information.

Levels of Curation – (Re)Appraisal

Repository Level	Object Level
Standard/Minimum/Maximum **ReAppraisal Periods** (Time)	ReAppraisal Period (Time) and Start (Date) for this specific object.
ReAppraisal Criteria	**Link** to repository level criteria in place when the object was ReAppraised
Outcomes. All potential outcomes of a ReAppraisal	History of reAppraisal outcomes for this specific digital object as part of provenance information.
Exceptions to ReAppraisal periods applied by the repository.	Exception to ReAppraisal periods for this specific object.

Other factors

- Community Served
- Technical
  - Formats served
  - Tech watch
- Semantic
- Quality Factors

Levels of Curation – Conclusion

- Furthering the levels of curation with metadata would:
  - Support repositories with a range of curation, preservation and retention activities
  - Allow researchers, funders and other stakeholders to better understand a repository's services
  - Highlight this information at both repository-level and the object-level
  - Support transparent descriptions of care over time
  - Allow for more machine-actionable evaluation and assessment

Further reading and chances for feedback

- CoreTrustSeal Standards and Certification Board. (2022). CoreTrustSeal Trustworthy Digital Repositories Requirements 2023-2025 Extended Guidance (V01.00). Zenodo. https://doi.org/10.5281/zenodo.7051096
- CoreTrustSeal Standards and Certification Board. (2024). Curation & Preservation Levels: CoreTrustSeal Position Paper. Zenodo. https://doi.org/10.5281/zenodo.11476980
- CoreTrustSeal Standards & Certification Board. (2024). Types of Repository: Entities, Responsibilities, Objects. CoreTrustSeal Board Discussion Paper (v01.00). Zenodo. https://doi.org/10.5281/zenodo.13133041
- L'Hours, H., Kleemola, M., & Recker, J. (2024). CoreTrustSeal Levels of Curation and Preservation: Implied Repository and Object Metadata Characteristics (v01.00). Zenodo. https://doi.org/10.5281/zenodo.12701324

CoreTrustSeal are starting to gather feedback for 2026-2028 requirements

Feedback welcomed via comments in this Google document

Any questions?